

From Radiation Bursts to QEC Syndromes

Simulation and Epicenter Localization



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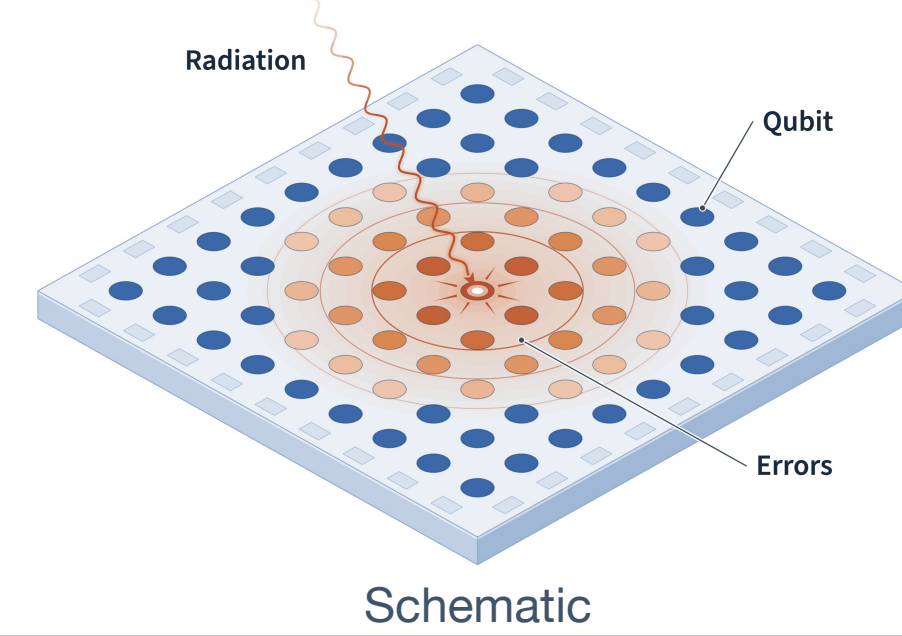
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Introduction

Can patterns in QEC detection records reveal where a radiation event started?

Background

Quantum error correction can be more challenging than in classical computers. Radiation can disturb several qubits at once, creating **correlated errors that are harder to correct**. Quantum error correction (QEC) uses repeated measurements to look for signs of errors. Comparing the results over time reveals unexpected changes, called detection events. These records **do not directly identify where the radiation event started**.



Our approach

We developed a **simulator** that connects radiation-like disturbances to QEC detection records. Using **simulated records with known event locations**, we trained a **machine-learning model, support vector regression (SVR)**, to estimate where each event began. We compare localization methods using **the same records** and examine how spatial and temporal information and qubit layout affect accuracy.

1 Radiation-to-syndrome pipeline

Event geometry and apparent propagation

Epicenter · shape · range · strength

Local source and quasiparticle dynamics [2]

$$\frac{dx_{qp,q}}{dt} = g_q(t) - \frac{sx_{qp,q}}{T_{trapping}} - \frac{rx_{qp,q}}{T_{recombination}}$$
$$\frac{1}{T_{1,q}(t)} = \frac{1}{T_{1,q}^{baseline}} + C_q x_{qp,q}(t)$$

Density change Generation Trapping Recombination

Relaxation rate Baseline relaxation QP-induced relaxation

Time-dependent relaxation → Pauli noise → Stim QEC

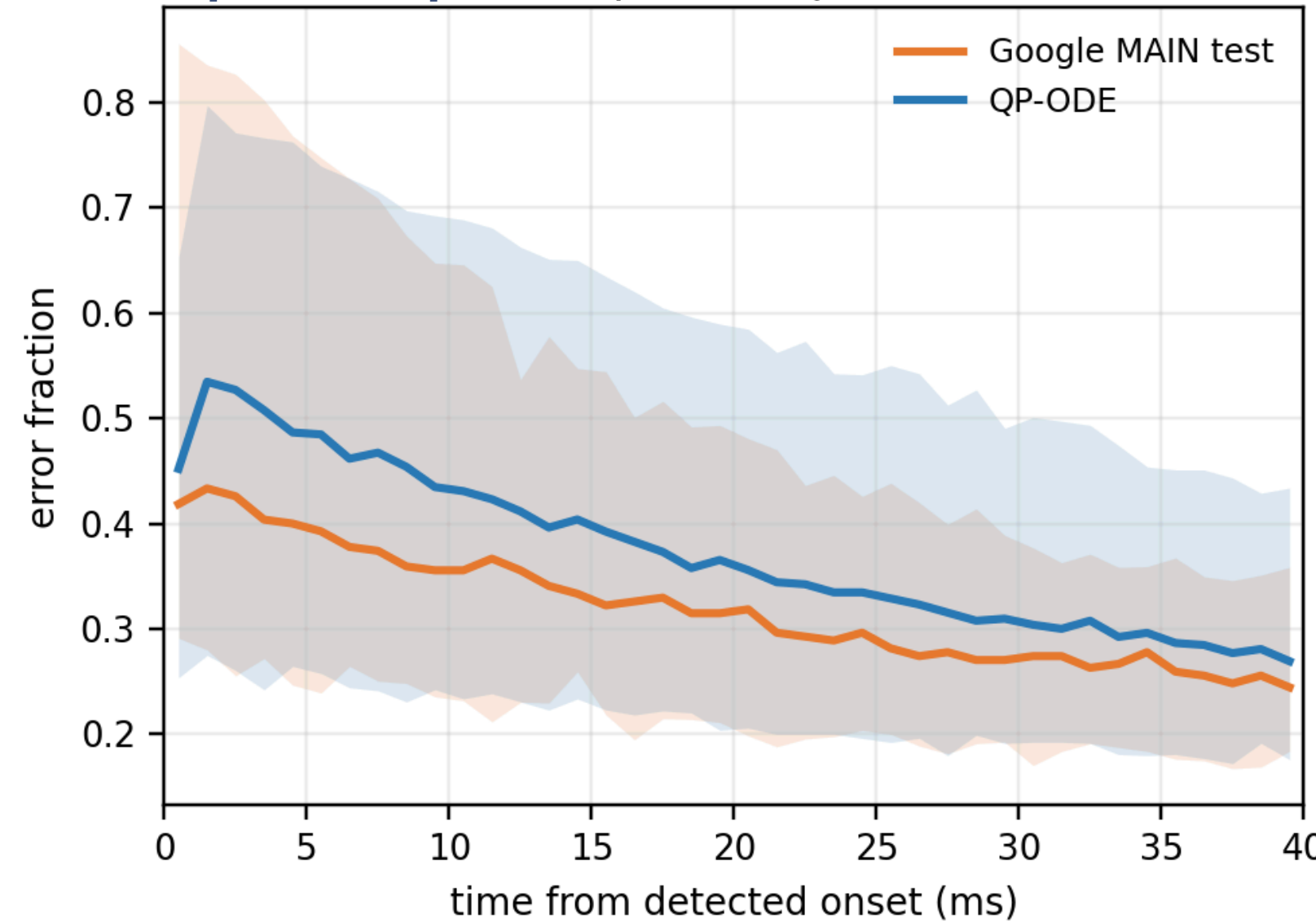
Detector records → estimated event coordinates

Detector records identify which checks respond and when. Simulated events with known origins provide data for training and evaluation.

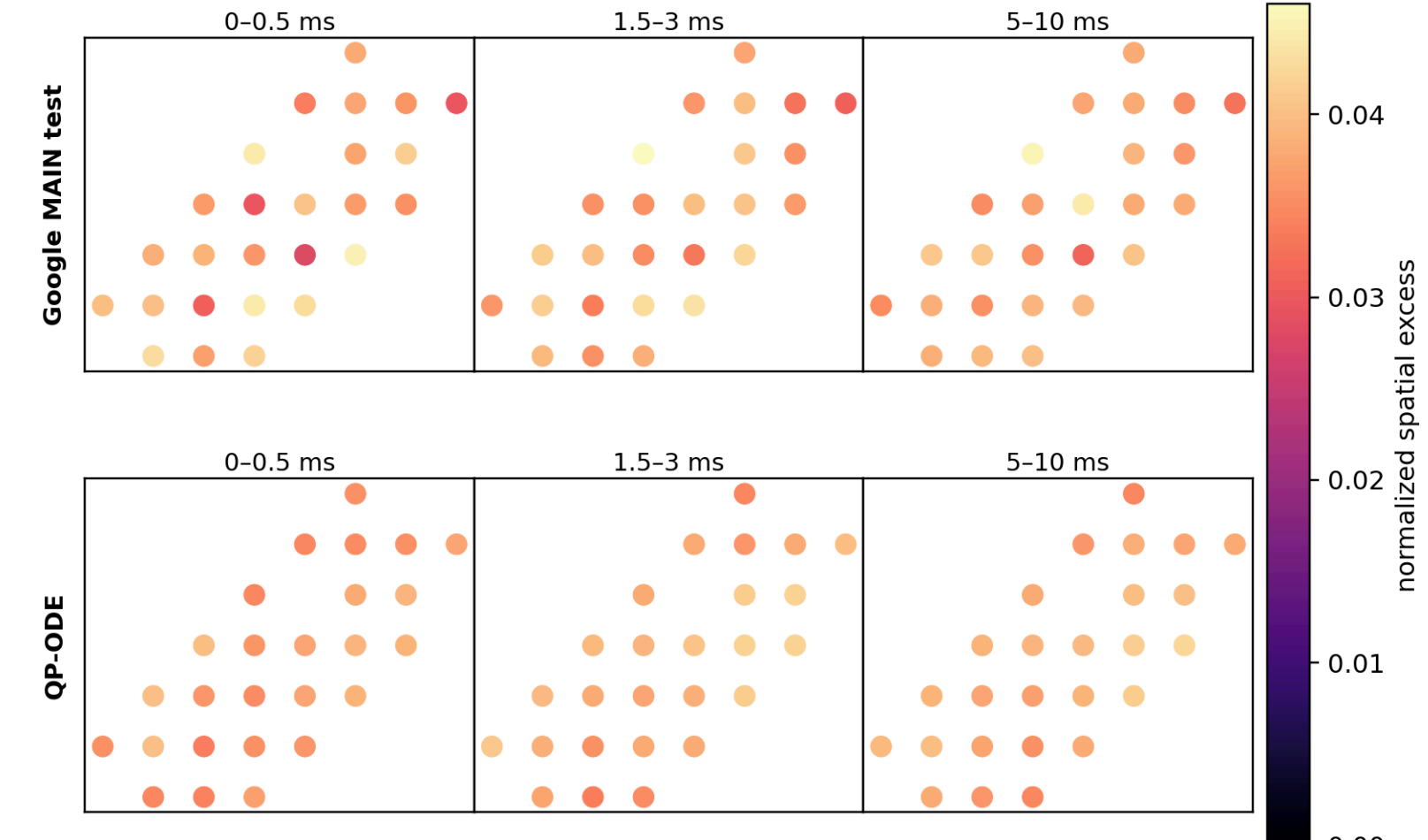
2 Measured temporal and spatial response

We compare simulated and measured relaxation responses [3] to evaluate the temporal and spatial behavior of the model.

A Temporal response



B Spatial response



Measured responses provide a reference for evaluating the simulator.

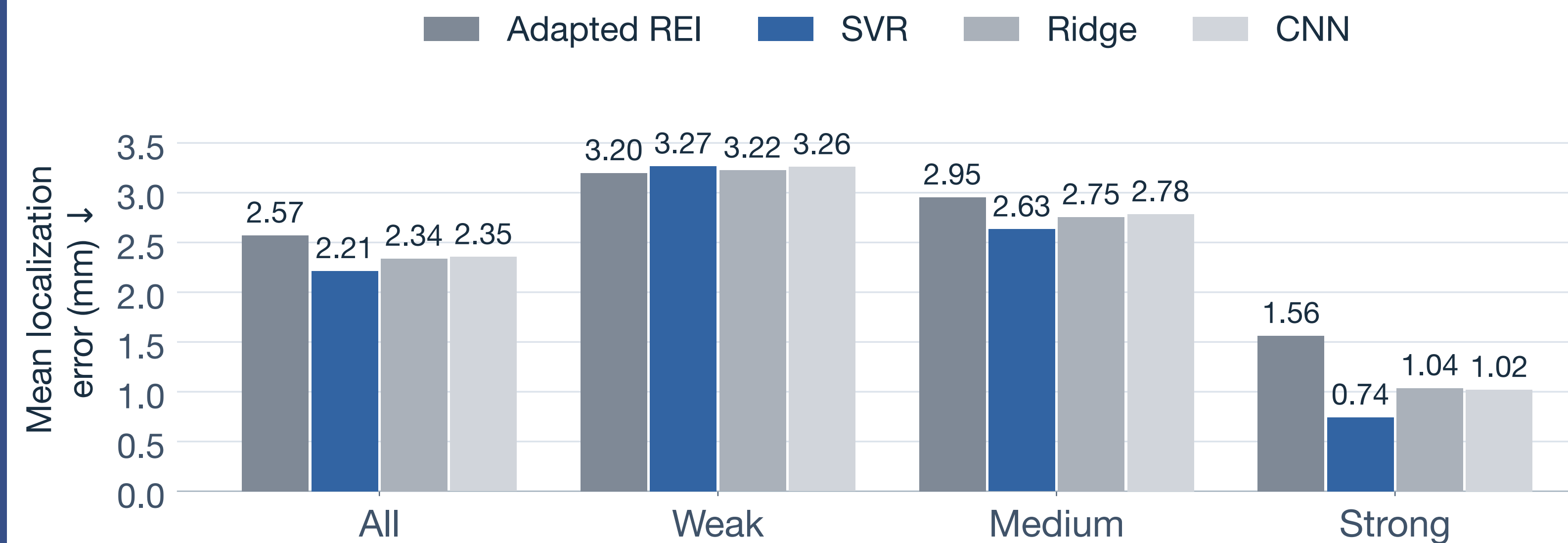
Main result

13.9% lower mean error

2.571 → 2.214 mm

Same ~4 ms simulated records · 540 physical test events

Adapted REI [1] → SVR

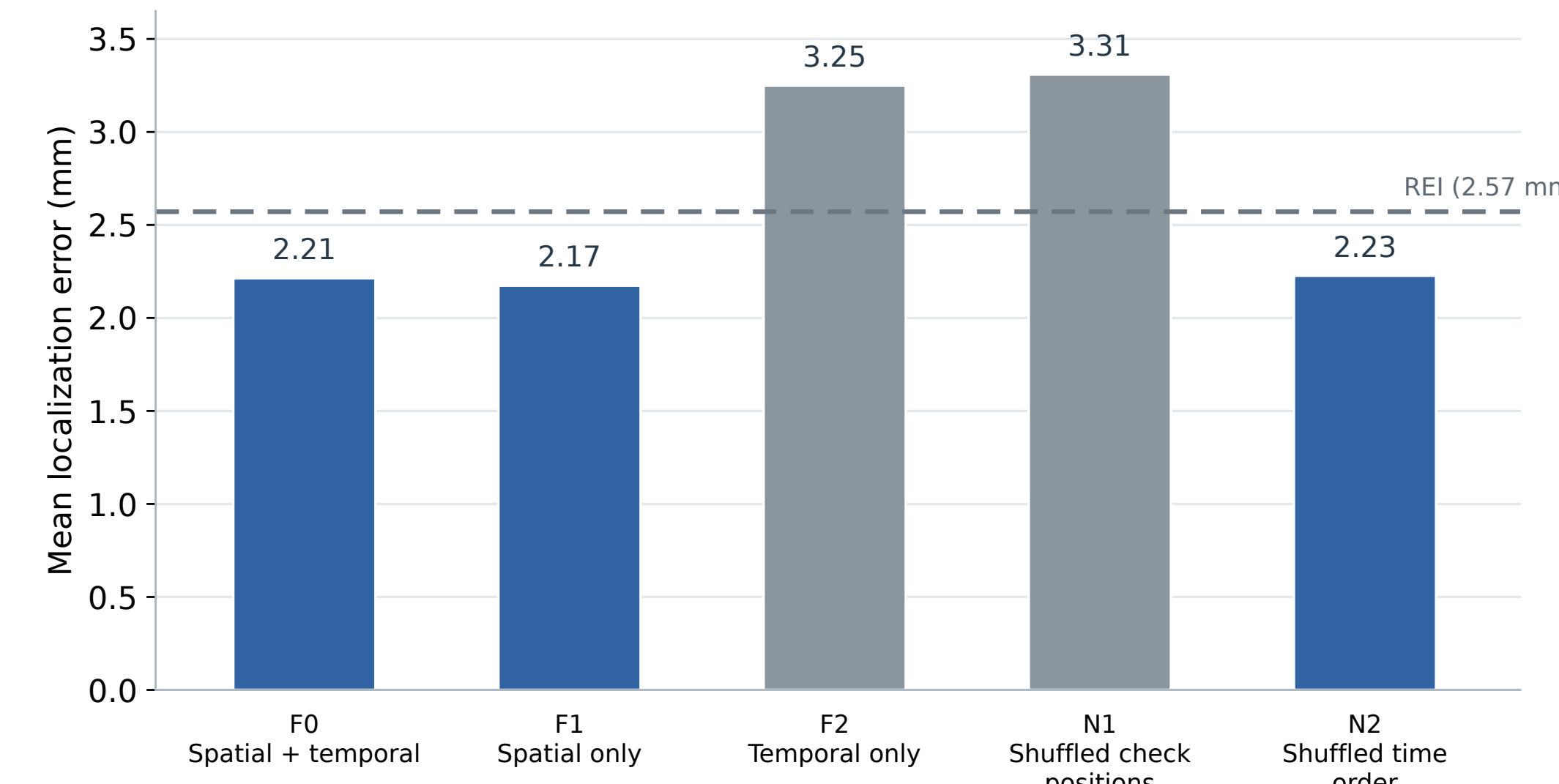


Gains are largest for medium and strong events, whose detector patterns provide information about event location.

Strong events within 1 mm: 23.9% → 77.8%

Spatial and temporal features

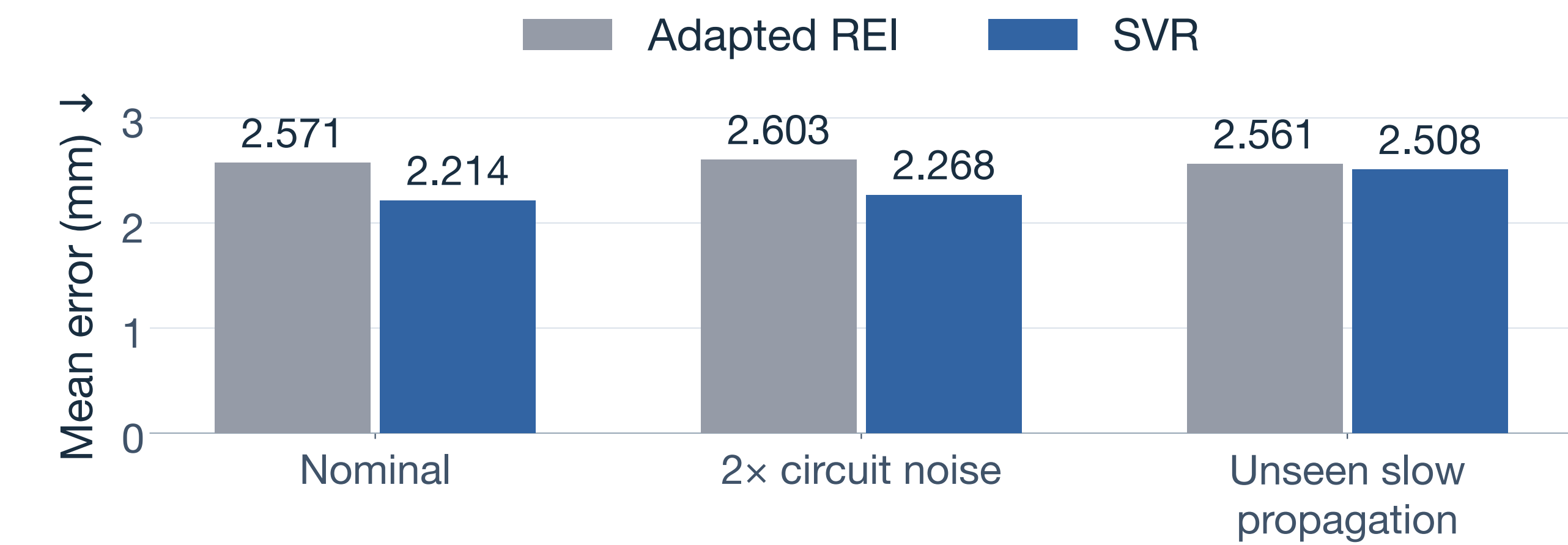
We remove or shuffle spatial and temporal information in the same records to assess its contribution to localization.



Spatial features alone retain similar accuracy. Removing spatial information or shuffling check positions increases localization error.

The spatial pattern of responding checks is a key localization cue.

4 Noise and propagation shift



SVR retains a lower mean localization error when circuit noise is doubled.

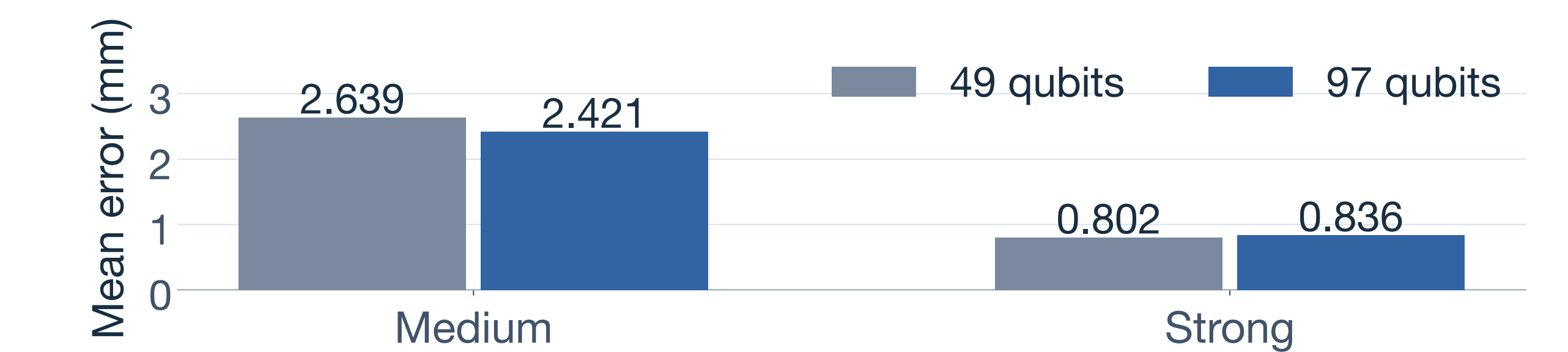
SVR also has a slightly lower mean error under slower propagation, suggesting some transfer across propagation conditions.

5 QEC layout as sensor geometry

We compare higher qubit density within a fixed area with wider spatial coverage at fixed spacing.

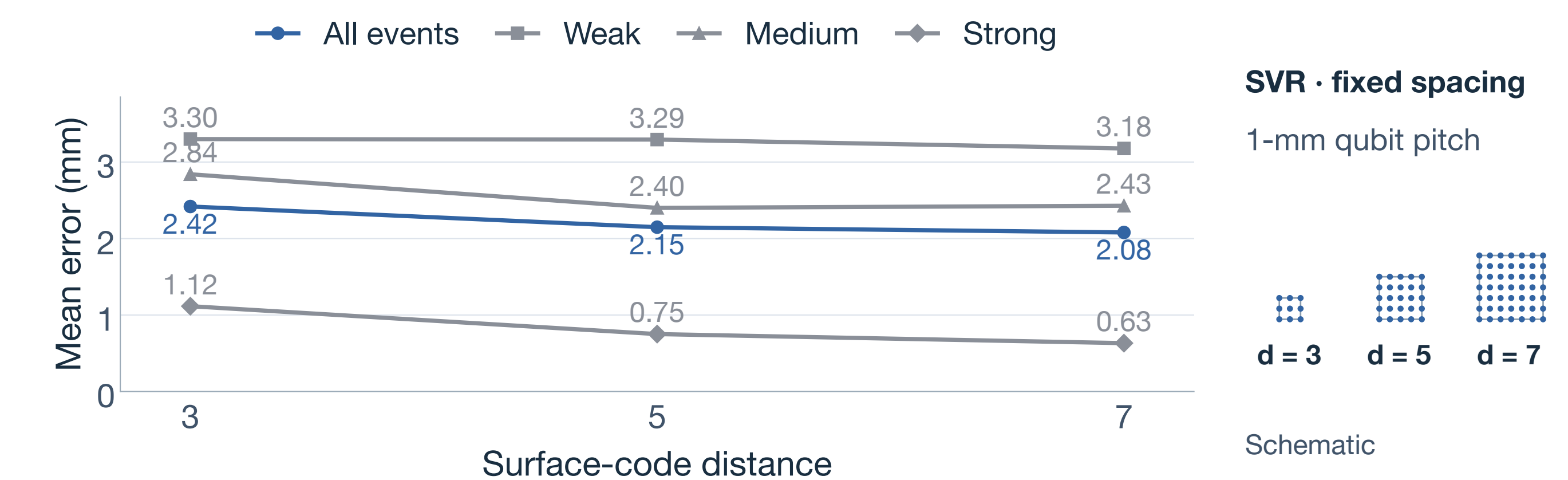
A Higher density, fixed area | ~4 ms

49 qubits / 24 checks → 97 / 48
Fixed area; SVR used for both layouts.



Higher qubit density reduces localization error for medium events at fixed area.

B Wider coverage: code-distance scaling



Wider coverage reduces the overall mean error. Strong-event error decreases across d = 3, 5, and 7.

Future work

Validate localization using hardware QEC records and investigate how location estimates can inform decoding.
Apply reservoir learning for event detection and epicenter localization.